

I0: Introduction and System Setup [READ BEFORE STARTING I1] ↕



Individual Assignment 0: Introduction and Setup

The first set of assignments are meant to expose you to the different aspects of the central team project that you will be completing for the rest of the semester, but on a smaller scale. This document contains important preliminary information about not only the project itself but the course as a whole.

Academic Honesty Policy:

The full academic honesty policy is available on Courseworks. It is mandatory that you read and understand the policies as they are written; you will be required to submit (in the first assignment) an academic honesty pledge and you will be required to keep this file in your repository throughout the duration of the entire individual project. Failure to include this file will result in an immediate 0 on your assignment. Some key highlights that are relevant to the individual assignments:

1. AI services of **ANY KIND** (ChatGPT, Copilot, Claudia, Gemini etc) are prohibited on these assignments
2. If you consult other sources, such as StackOverflow, you must document the instance and how they affected your submission in a file called "citations.txt"
3. Under no circumstances are you allowed to use any source as described in (2) to acquire 'cheat code'; refresh yourself of this definition from Professor Jae's policies: ([Academic Honesty Policies \(columbia.edu\)](https://www.cs.columbia.edu/~jae/honesty.html) (<https://www.cs.columbia.edu/~jae/honesty.html>))
4. There is no collaboration between yourself and other students allowed for these individual assignments; group work will come later.

Course Prerequisites:

COMS 3157: Advanced Programming (or equivalent) is a very much enforced prerequisite for this course. Most of the following should also apply to you

1. Be able to read code in both Java and C - most code examples used in class will be in Java or C.
2. Be able to write code fluently in at least one of Java or C/C++ at the full-stack or backend level, not just frontend. The team project will be a server or service (NOT an app) implemented in your team's choice of Java or C++. The course will not teach these or any other languages, you are required to

already know Java or C/C++. **Python is not allowed (except for very limited use with AI components optionally included in team projects) and students who only know Python should not take this course.**

3. Be able to maintain your own personal codebase(s) on GitHub or similar version control repository.
4. Know how to search for and read (or watch) documentation and tutorials for software development tools, frameworks, APIs, libraries, etc. on your own, without instruction or assistance from the teaching staff.
5. You have read chapter 1 of the "Software Engineering at Google" book at <https://abseil.io/resources/swe-book/html/ch01.html>  (<https://abseil.io/resources/swe-book/html/ch01.html>) and the "coping with hitchhikers and couch potatoes on teams" article at <https://ieeexplore-ieee-org.ezproxy.cul.columbia.edu/stamp/stamp.jsp?tp=&arnumber=9625655> (<https://ieeexplore-ieee-org.ezproxy.cul.columbia.edu/stamp/stamp.jsp?tp=&arnumber=9625655>) and still want to take this class.

If you have any questions about these prerequisites please contact the Head IA at the following address: gcn2106+4156@columbia.edu (<mailto:gcn2106+4156@columbia.edu>) or make a post on Ed Discussion

System Setup For Individual Project:

There are two versions of this project available for you to work on. The first version was developed using Java 17 and Maven 3.9.5 so you will need to have those installed along with an IDE (we recommend IntelliJ).

The second was developed using C++ and the setup is a little more involved. C++14 was used as well as cmake version 3.27.9 but any version higher than 3.15 should suffice. You will also need to download two external libraries: **boost**  (<https://www.boost.org/users/download/>), **crow**  (<https://crowcpp.org/master/>), and you will need to install **asio**  (<https://think-async.com/Asio/>) as well (which is typically done via brew install or windows equivalent). Inside the project there is an external libraries folder where you should put the unzipped folders for boost and crow. At the root level of the repository there is a file called: "CMakeLists.txt" you'll need to make sure of the following things:

1. Line 5 should list the names of the locations of the boost and crow folders
2. Lines 33 & 55 contain the location of your asio installation

You will also need to use VS Code for this as well as install cpplint (pip install cpplint should do the trick) There are several extensions on VS Code for C++ that you should install, you can also install your static analysis tool (cppcheck) via this route as well or the standard brew install will do. To make your project change your present working directory to the build folder and then run the following commands:

1. cmake ..
2. make

Assuming everything goes well, you'll have two executables to run, one which runs the program and another to run the test suite. You will need to find a way to do coverage analysis. Please note that the setup instructions are slightly different for windows as there is no native make command etc so please ensure you have some way to continue via windows if that is your platform of choice (using mingw is a common route but this is up to you)

The GoogleTest version changes frequently and the fetch URL may not work due to this, if this happens you can either find the latest release and replace it or the one listed at [this link](https://google.github.io/googletest/quickstart-cmake.html)  (<https://google.github.io/googletest/quickstart-cmake.html>) should do the trick.

The general point is this: the Java version works seamlessly across both Windows and MacOS, and while the C++ version is able to run on Windows, the setup is much more complex and requires individual effort to get working, in the event you know both Java and C++ and are on Windows I strongly recommend using the Java version

Definition of a 'Substantial Pull Request'

Throughout the individual project you will be required to make a minimum number of 'substantial pull requests'. Within the Software Engineering industry a pull request (PR) typically represents a unit of work. What this means in terms of setup is that when you fork over the repository, you should do the following things:

1. [OPTIONAL] Add a branch protection rule for the main branch
 - [Here's some documentation](https://docs.github.com/en/repositories/configuring-branches-and-merges-in-your-repository/managing-protected-branches/managing-a-branch-protection-rule)  (<https://docs.github.com/en/repositories/configuring-branches-and-merges-in-your-repository/managing-protected-branches/managing-a-branch-protection-rule>)
 - You only need to ensure that a pull request is required for merging
2. For each significant contribution or unit of work in the assignment, you should create a separate branch. You will then check out this branch and make your changes. Once your work is complete, push the branch to the remote repository and create a pull request to associate your changes with that branch.
 - You must select your own forked repository as the base repository of the PR. No credit will be awarded for PRs to other repos.

Each pull request can contain multiple commits, so if you are a person who likes to commit after every tiny change or a person who makes a single large commit this will work out for you. Considering that pull requests can vary in size, we define a 'substantial pull request' as the following:

A pull request that contains any number of commits such that the following is true:

1. Changes included within are representative of solid progress towards goals outlined in the assignment
- Ex: Adding a dependency wouldn't satisfy the PR requirement, but adding a dependency AND some code that helps achieve assignment goals would satisfy the substantial PR requirement.
2. Does not indicate any brute force attempt to inflate the count
- Ex: Say you have a pull request that satisfies the above but then you make another one that removes the changes from the first one. This would not count as a substantial pull request.

The minimum number of 'substantial pull requests' provided for each assignment was determined by those who tested the assignment beforehand and provided feedback. If you prefer to have a copy of this document, you may access a pdf version in the Files > Individual Assignments > Specs > I0.pdf

Points

0

Submitting

Nothing

Due	For	Available from	Until
-	Everyone	Sep 3, 2024 at 12am	-

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